Preface

The purpose of this publication, and the accompanying software called *Industrial Control in Practice (Issued versions named "ICIP2PRO" for DOS and "ICT32" for Windows)* is for training in industrial process control.

Experience shows that most graduate engineers have a sound knowledge of the mathematical aspects of process control. Nevertheless, when it comes to the practical understanding of industrial process control, there is often a problem in converting this theoretical knowledge into a practical understanding of control concepts and problems.

This publication, with its simulation software and associated training workshop, is intended to fill this gap.

It is not intended to add another book to the vast number of existing books, covering process control theory. The book *"Feedback and Control Systems"* is recommended as a good theoretical and mathematical complement to this publication. This document provides a practical understanding of control concepts as well as enabling the reader to gain a correct understanding of control theory.

Although the mathematical side is kept to a minimum, a basic understanding of engineering concepts and a general understanding of algebra and calculus is required in order to obtain a full understanding of this publication

Together with this documentation, software for realistic process simulation and control is provided to allow for extensive practical exercises. These realistic process simulations provide a tool necessary to gain greater practical and theoretical knowledge and a clearer perception into control concepts.

Practical experience and confidence can be achieved if full use is made of this software. The principals of industrial process control concepts and the associated pitfalls are explained in an easy to understand manner.

For example, perfect control of a valve (according to process control theory) could result in excessive valve movement. This however would most likely result in frequent valve failure with associated production losses.

There is a degree of emphasis on the internal calculation of control algorithms in digital computers. The purpose of this is to provide a wider view of the use and modification of computer calculated algorithms (incremental algorithms).

The first automatic control system known was the Fly-ball governor installed on Watt's steam engine in 1775 to regulate the steam rate. It was nearly a

century later that the first mathematical model of the Fly-ball governor was prepared by James Clerk Maxwell. This illustrates a common practice in the development of process control, using a system before fully understanding exactly why and how it does the job. The spreading use of steam boilers resulted in the introduction of other automatic control systems, such as steam pressure regulators and the first multiple element boiler feedwater systems. Again, the applications came before the theory.

The first general theory of automatic control, written by Nyquist, only appeared in 1932.

Today, automatic control is an increasingly important part of the capital outlay in industry. The primary difficulty encountered with process control is in applying well-defined mathematical theories to day-to-day industrial applications, and translating ideal models to the frequently far from ideal real-world scenario.

Process control has a number of significant advantages. As always, the primary factor in any operation is cost. The use of process control in a system enables the maximum profitability to be derived. Other advantages are that automatic control results in increased plant flexibility, reduced maintenance, and in stable and safe operation of the plant.

It also allows operators to more closely approach optimum operation of the process. As the degree of automatic control is increased, so too do the related advantages become more significant.

Further improvements in process control are attained by model based control and ultimately by optimization.

Optimization applications can be installed when the plant is stable, is being operated safely, and has tight quality control. The benefits of optimization are an improvement in product yield and quality, reduction in energy consumption, and a move to optimum operation of the process. It is possible to track optimum operation to maintain the maximum profitability of the process.